

Studies of ECAL Timing Reconstruction

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Abstract

One of the most exciting elements of the future CMS detector anticipated for the High Luminosity LHC (HL-LHC) era, will be the incorporation of precision timing. A combination of new sub-detectors (MTD, HGCAL) and dedicated upgrades to existing detectors, such as the ECAL, will facilitate 4D reconstruction of events. 4D reconstruction methods will provide new tools for both pile-up suppression and a range of new experimental techniques to probing physics signatures and long-lived particles. To facilitate the incorporation of precision timing in the HL-LHC era, studies of the ECAL timing reconstruction were conducted and a new time reconstruction, the Cross Correlation Time Algorithm (CC), was developed. The CC algorithm mitigates the effects of out-of-time pileup on the ECAL time reconstruction by leveraging the existing, pile-up aware, ECAL energy reconstruction method. In order to measure the performance of this new algorithm, refinements to existing methods to determine timing resolution and new methods to determine approximate GEN truth times were developed. The CC algorithm is shown to improve the single crystal resolution by ~ 30 ps over the Run II default method and is expected to achieve analysis relevant E/Gamma object resolutions of ~ 100 ps, compared to ~ 300 ps typical in Run II. With the utilization of the CC algorithm we expect comparable, if not improved, resolutions in Run III.

Preliminary version

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1 Introduction

After Run-3 of the LHC a significant upgrade is planned for the HL-LHC [1] period of operation. Expected instantaneous luminosities in the HL-LHC era will be a factor of five times larger than the current LHC nominal conditions. While these more intense conditions will enable the LHC experiments to accumulate more than an order of magnitude larger dataset than all of the preceding LHC runs, it will also exacerbate the experimental challenge of the multiple pile-up interactions appearing in each event. To mitigate degradation of LHC event reconstruction performance resulting from the potentially hundreds of pile-up (PU) interactions expected in an HL-LHC event, both CMS [2] and ATLAS [3] are developing methods to incorporate precision timing measurements into the detectors, significantly expanding the capabilities of the experiments. The inclusion of precision timing measurements through a combination of new sub-detectors (MTD, HGCal) and dedicated upgrades to existing detectors, such as the CMS electromagnetic calorimeter (ECAL). This will also facilitate the use of space and time (4D) in the reconstruction of events. 4D reconstruction methods will provide new tools for both pile-up suppression and a range of new experimental techniques to probing physics signatures and long-lived particles. The ECAL has set a goal of upgrading its systems to achieve ~ 30 ps time resolution in the HL-LHC era [4].

The ECAL is a hermetic homogeneous calorimeter made of 75 848 lead tungstate (PbWO_4) scintillating crystals arranged in a barrel (EB) and two endcap (EE) sections [5]. Lead tungstate has a fast scintillation response and is resistant to radiation. These crystals have a high density of 8.3 g/cm^3 , a short radiation length of $X_0 = 0.89 \text{ cm}$, and a small Molière radius of $R_M = 2.0 \text{ cm}$, allowing for a highly granular, compact detector. Each individual crystal is a truncated pyramid, with a lateral size comparable to R_M and a length of $25.8 X_0$ for the barrel, and $24.7 X_0$ for the endcaps. When an electromagnetic particle interacts with a crystal, and thereby deposits energy within the crystal, a burst of scintillating light is emitted. The scintillation decay time of the crystals is comparable to the LHC bunch crossing interval of 25 ns in which 80% of the light is emitted. The intensity of the scintillating light bursts are measured with avalanche photodiodes (APD) in the barrel and vacuum phototriodes (VPT) in the endcaps [6].

While the primary purpose of the ECAL is to measure energy deposited in the calorimeter by incident photons and electrons, the combination of the scintillation timescale of PbWO_4 crystals, the electronic pulse shaping, and the sampling rate, allow excellent time resolution to be obtained. The time resolution achievable in the ECAL directly impacts the energy resolution and plays a significant role in many important physics measurements. Currently, time measurements allow the rejection of backgrounds with a broad time distribution such as those arising from cosmic rays, beam halo muons, electronic noise, secondary interactions within the material of the support structures and electronics of the detectors, and out-of-time proton-proton interactions [4]. The timing resolution of the ECAL also directly impacts the sensitivity of many types of searches, such as those involving long lived particle (LLP) decays. When a particle does not decay promptly, the arrival time at the ECAL of particles resulting from the decay of this particle can be measurably later than those coming from the decay of short lived particles.

The time reconstruction is sensitive to the scintillating light pulses in a crystal that arise from particles produced in collisions other than those of the particles produced in the particular collision that are to be reconstructed. Such extraneous collisions present in a reconstruction are called pile-up (PU) collisions. When these PU collisions originate in the bunches of colliding protons proceeding or following the bunch of colliding protons that produced the reconstructed interactions, this is referred to as out-of-time pile-up (OOT-PU). As the LHC moves to higher luminosity in the HL-LHC era, the effects of OOT-PU in events become a significant issue in many areas. This will not only directly degrade the time resolution, but since the time measurement is used to reject backgrounds arising from OOT-PU, the resolution of many physics measurements can be degraded. A significant step to prepare for operations in high OOT-PU environments will be to study methods to mitigate the effects of OOT-PU on the ECAL timing reconstruction. A timing reconstruction method that is OOT-PU aware will result in an improved time resolution that will not only improve the precision of many current physics measurements, it will also provide a foundation to meet the goal of achieving a ~ 30 ps time resolution in the ECAL. Such a timing method will support the incorporation of precision timing in the HL-LHC era, and will also allow us to begin to explore in Run III some of the possibilities of 4D reconstruction methods facilitated by precision timing measurements in the HL-LHC era.

In this paper we will discuss time reconstruction methods in Section 2. First, in Section 2.1 we will review the preceding time reconstruction, the ratio method, and the challenges faced with this reconstruction. We will then present, in Section 2.2 in the details of new the time reconstruction, the Cross Correlation method. In this section we will also discuss the nature of, and the setting of the parameters used to reject, anomalous reconstructed hits, called “spikes”, in Section 2.2.2. We will also discuss the sensitivity of the new time reconstruction on out-of-time amplitudes derived in the energy reconstructed in Section 2.2.1. Finally, the performance of the Cross Correlation

time reconstruction in data is presented in Section 3. Finally, this paper is summarized in Section 4.

2 ECAL Time Reconstruction Algorithms

The ECAL measures both the energy deposited in the calorimeter by incident particles, and the time at which that energy was deposited. In both cases these measurements are extrapolated from the scintillating light pulse produced when the incident particle interacts with the PbWO_4 crystal. When a particle is incident in a PbWO_4 crystal, it will produce an electromagnetic shower of electron - positron pairs and photons. The showering particles will dissipate their energy through ionization, leading to the excitation of the scintillating centers of the PbWO_4 crystal. The scintillating centers will de-excite producing a pulse of scintillating light at a specific wavelength that is proportional to the energy of the original, incident particle. The “signal” pulse is amplified and shaped by the readout electronics, thereby defining the shape of the pulse [7]. This shape is constant for all sampled pulses in a particular crystal (or channel) and is very well known. Ten consecutive amplitude samples of this “signal” pulse are recorded. In Figure 1 the time structure of a typical signal pulse shape as measured after amplification (solid line) is shown. The normalized amplitude of the pulse, A/A_{max} , is a function of the time difference $T-T_{max}$, where T_{max} is defined as the time when the pulse reaches its maximum value. The dots are the ten discrete samples of the pulse amplitude.

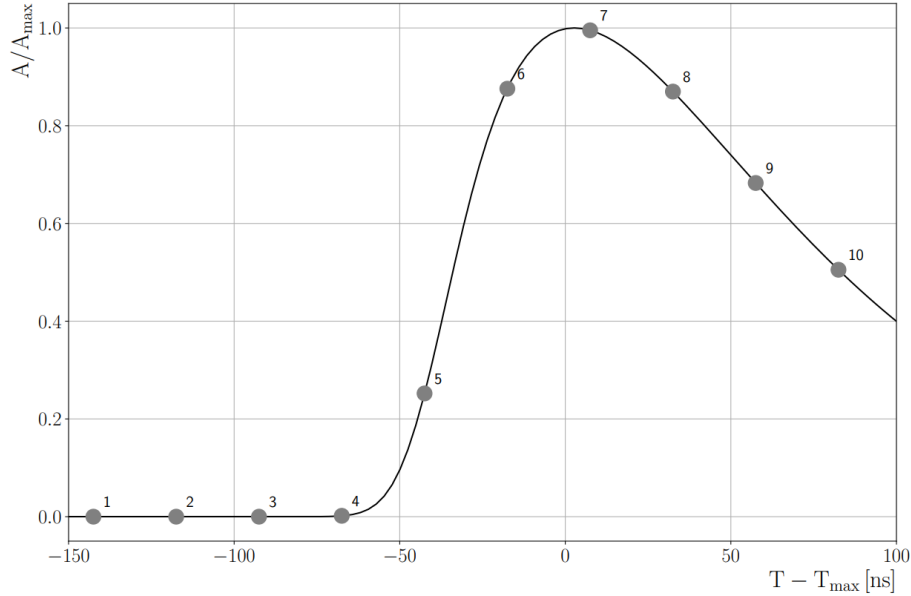


Figure 1: The typical pulse shape measured in the ECAL, as a function of the difference between the time (T) of the ADC sample and the time (T_{max}) of the maximum of the pulse and the amplitude of the pulse (A) normalized by the maximum amplitude of the pulse (A_{max}). The dots indicate ten discrete samples of the pulse as taken from a single event with the pedestal (the amplitude measured when no pulse is present) subtracted.

During LHC Run II, from 2015 through 2018, the instantaneous luminosity, and therefore the number of interactions in each collision event, or bunch crossing (BX), increased by a factor of two compared to Run I. These conditions placed much more stringent requirements on the ECAL pulse amplitude reconstruction algorithm. In more detail, the LHC beams are composed of bunches of protons that are localized in buckets dictated by the RF cavity frequency of the LHC. When two bunches of protons cross in the center of the detector, thus producing proton-proton collisions, it is referred to as a bunch crossing. With the increase in luminosity in Run II, the number of inelastic collisions occurring during a BX, referred to as pile-up (PU), at the LHC increased to an average of forty. This results in an increase in the amount of particles interactions recorded during a single BXing. In addition, the spacing between BXs was reduced from 50 ns to 25 ns. With this shortened spacing between BXings it becomes more likely that interactions with particles originating in collisions that occurred in the previous BXing are recorded in the current BXing. This is referred to as out of time pile-up (OOT-PU). The net result is that the observed pulse shape of a signal in the ECAL is a superposition of a nominal, or in-time, pulse that is of interest and multiple, non-negligible, out-of-time (OOT) pulses that originate in both PU and OOT-PU interactions. This is demonstrated in Figure 2 with a single OOT pulse for clarity. To account for the increase in both in-time PU and OOT-PU, a template fit with multiple components, termed “multi-fit”, was implemented for the ECAL energy

86 reconstruction.

$$\chi^2 = \sum_{i=1}^N \frac{(\sum_{j=1}^M A_j p_{ij} - S_i)^2}{\sigma_{S_i}^2} \quad (1)$$

87 The multi-fit algorithm estimates the in-time signal amplitude and up to nine OOT amplitudes through the mini-
 88 mization of the χ^2 given in Equation 1. In this equation, A_j are the amplitudes, j , of up to ten, $M = 10$, events.
 89 The pulse templates p_{ij} for each BX_j have the same characteristic shape, but are shifted in time by multiples of
 90 25 ns (the frequency at which the pulse amplitudes are sampled) within a range of -5 to $+4$ BXs around the time
 91 of the in-time signal at BX zero. The pulse shapes for each crystal were measured from low pileup pp collision
 92 data recorded by CMS at the beginning of 2015 and a periodically updated library of these pulse shape templates
 93 is maintained. Noise generated by electronics associated with the crystal readout chain is characterized by $\sigma_{S_i}^2$
 94 and is measured from dedicated pedestal runs. The technique of Non-Negative-Least-Squares is used to perform
 95 the χ^2 minimization of Equation 1 with the constraint that the fitted amplitudes are all positive [8]. Information
 96 reconstructed for a pulse, resultant from a particle that has hit a crystal, is referred to as a reconstructed hit or a
 97 rechit. While the process of determining the energy for a rechit involves some technical details not discussed here,
 98 it is sufficient for our purpose to say that the pulse amplitude determined by the multi-fit algorithm is multiplied
 99 by a periodically calibrated constant, per crystal, to determine the energy.

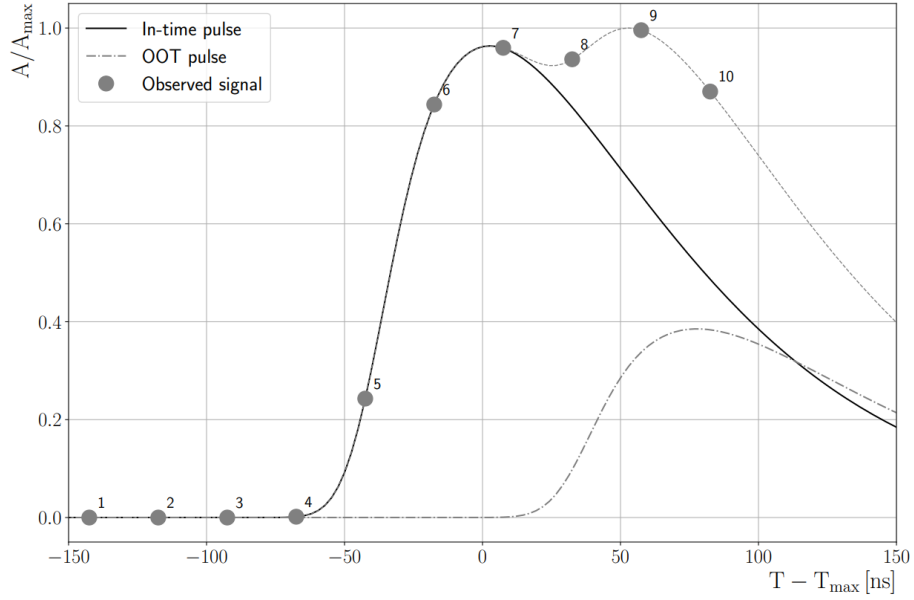


Figure 2: The fitted pulse for a simulated event with twenty average pileup events and 25 ns bunch spacing as a function of the difference between the time (T) of the ADC sample and the time (T_{max}) of the maximum of the pulse and the amplitude of the pulse (A) normalized by the maximum amplitude of the pulse (A_{max}). The dots represent the 10 discrete samples of the measured pulse shape (dashed line) that is a superposition of an in-time pulse (solid line) and an OOT pulse (dashed line).

100

101 The ECAL time reconstruction developed for Run 1 was utilized through the start of Run 3. With this time re-
 102 construction it was found that the fast time constants of $PbWO_4$ scintillation and the response of the readout
 103 electronics allows a time resolution on the order of 100's of picoseconds for isolated interactions [9]. Residual
 104 channel-to-channel time offsets are corrected with appropriate constants derived from data. This timing informa-
 105 tion, combined with topological information of the energy deposits, is exploited at reconstruction level to reject
 106 signals inconsistent with the emission of scintillation light by particles produced in proton-proton collision events.
 107 These spurious signals include those arising from direct ionization in the APD sensitive region (spikes) as well as
 108 signals arising from beam halo muons, cosmic rays, electronic noise, detector material interactions, and OOT-PU.
 109 In Run 1 it was found that, for the conditions prevalent at that time, the residual contamination of these spurious
 110 deposits has a negligible impact on reconstruction [10]. The signal arrival time for a rechit is reconstructed by

comparing the relative phase of the sample pulse amplitudes to the expected pulse shape. In originally developed algorithm used the ratios of consecutive amplitude samples to reconstruct the time. This method did not take into account the challenges presented by the increasing PU present in Run 2 and Run 3 to the time reconstruction. The ratio method algorithm, and the challenges it faced, is discussed in Section 2.1. In order to account for these changes a new time reconstruction, the cross correlation method (CC), was developed during Run 2. The CC time reconstruction will be presented in Section 2.2.

2.1 Ratio Time Reconstruction

As for the energy reconstruction, in the ratio method the time of a rechit in an ECAL crystal is estimated from the ten consecutively sampled pulse amplitudes that are recorded every 25 ns. The ratio of any two consecutive amplitudes, sampled from a signal with the expected pulse shape, uniquely determines the timing of an isolated pulse measured in a particular crystal, relative to the LHC beam clock. In Figure 3, the ratio, R , of two consecutively measured amplitudes, $A(T)$ and $A(T + 25ns)$, is shown as a function of the time difference $T - T_{max}$, where T_{max} is defined as the time when the pulse reaches its maximum value. The first three measured pulse heights, $A(1)$, $A(2)$, and $A(3)$ are typically used to measure the pedestal values (measured amplitude corresponding to “zero”) for the pulse. Ratios for $R(A(4)/A(5))$ through $R(A(7)/A(8))$ are used to estimate the pulse timing. Since the remaining ratios arising from the tail of the pulse are insensitive to the pulse timing, they are not utilized. The rechit time for a pulse in a crystal is determined from the weighted averages of three consecutive values of $R(A(4)/A(5))$ through $R(A(7)/A(8))$.

The clock distribution system has its own delays relative to the actual collision time and these delays vary from crystal to crystal. To compensate for this the timing is calibrated using actual prompt relativistic particles so that the mean reconstructed time, per channel, is defined to be zero [11].

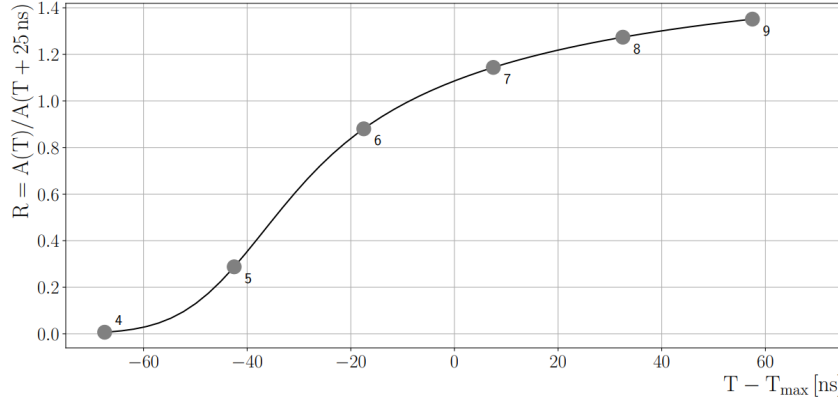


Figure 3: The ratio, R , of two consecutively measured amplitudes, $A(T)$ and $A(T + 25ns)$, as a function of the time difference $T - T_{max}$, where T_{max} is the time when the pulse reaches its maximum value. The first three amplitudes, $A(1)$, $A(2)$, and $A(3)$, typically used to measure the pedestals values for the pulse, are not used. The rechit time is determined from the weighted averages of three consecutive value of $R(A(4)/A(5))$ through $R(A(7)/A(8))$.

$$t_{ECAL} = \sum_t \frac{t_{ECAL}^i}{\sigma_i^2} / \sum_t \frac{1}{\sigma_i^2} \quad (2)$$

The time of arrival for a reconstructed particle at the ECAL, t_{ECAL} in Equation 2, is calculated based on a weighted sum of the reconstructed times in each of the ECAL crystals that compromises the cluster of crystals associated with the reconstructed particle. In this equation t_{ECAL}^i is the pulse time in crystal i . σ_i is the estimated time resolution of the signal pulse in crystal i . The time resolution weights σ_i^2 are estimated based on the intrinsic time resolution of the ECAL crystal sensors [9].

The ratio method assumes that the sampled signal pulse is well isolated in time. In the higher PU environments of Run 2 and Run 3 this is no longer a good assumption. As was previously noted, for Run 2 and beyond, the observed pulse shape of a signal in an ECAL channel is a superposition of a nominal, or in-time, pulse that is of interest and multiple, non-negligible, out-of-time (OOT) pulses. This situation is illustrated in Figure 2 with a single OOT pulse

for clarity. The resultant superposition of multiple pulses into a single sampled pulse shape is not well modeled by the expected pulse shape as measured in the pulse shape templates. In the case where the “sampled” pulse shape differs significantly from the pulse shape template, the ratio of any two consecutive amplitudes will not uniquely determine the timing of the pulse measured in a crystal.

2.2 Cross Correlation Time Algorithm

The cross correlation method utilizes the ten fitted pulse amplitudes, A_i^{mf} , derived from the multi-fit method for the energy reconstruction. The actual pulse shape, $H(bx)$, defined by the pulse amplitudes A_i^{mf} can be considered to be a linear combination of the nine OOT pulses and the nominal, in-time, pulse of the rehit in a crystal. It is reasonable to say that each of these pulses will arise from a particle incident in the crystal and will therefore exhibit the expected, well-known, characteristic pulse shape produced by a scintillating light burst. It is not reasonable to say that it is known exactly at what location on this expected pulse shape that a particular amplitude, A_i^{mf} , was measured.

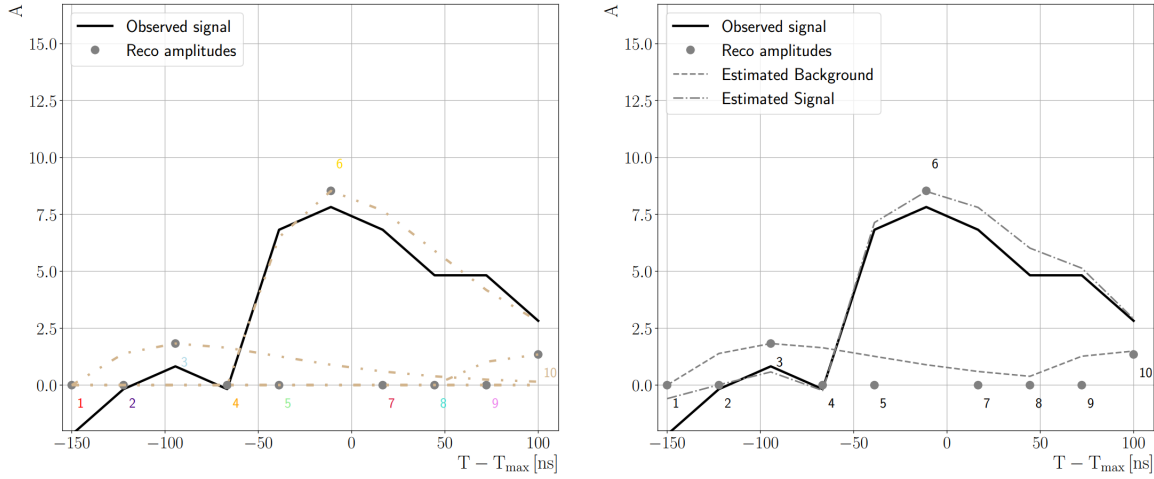


Figure 4: An illustration of the OOT amplitude correction for the proposed time reconstruction method. In both plots the black line is the measured signal, $H(bx)$ and the black dots are the reconstructed amplitudes from the multi-fit, A_i^{mf} . In the plot on the left the double dotted dashed lines are the nine OOT pulses, $F_i^{OOT}(A_i^{mf}, t)$, including the nominal, or in-time, pulse given by amplitude 6, that are all reconstructed from A_i^{mf} . In the plot on the right the dashed line is the background pulse shape, $F_{bg}(bx, t)$, that is the linear combination of the OOT amplitudes. The dotted dashed line is the recovered signal, $A^{sig}(bx, t)$, that is found by subtracting the background pulse shape from the measured signal.

A continuous function describing the characteristic pulse shape is derived with an analytic description that is fit, channel by channel, to the templates of the known characteristic pulse shapes. The amplitudes, A_i^{mf} , and this analytic description of the characteristic pulse shape is utilized to define a function, $F_i^{OOT}(A_i^{mf}, t)$, that represents the amplitude and time dependence of the shape of the OOT pulses. The sum, or superposition, of the OOT pulses then can be used to define a background pulse shape as given in Equation 3.

$$F_{bg}(bx, t) = \sum F_i^{OOT}(A_i^{mf}, t) \quad (3)$$

As shown in Figure 4, by subtracting $F_{bg}(bx, t)$ from $H(bx)$ we will recover the amplitudes for the nominal pulse which should have the characteristic pulse shape of a scintillating light burst if the correct time parameter for $F_i^{OOT}(A_i^{mf}, t)$ is chosen. Using this approach a signal function, $A^{sig}(bx, t)$, is defined in Equation 4 that gives the pulse shape of the in-time, nominal, pulse. This process is referred to as the OOT amplitude correction.

$$A^{sig}(bx, t) = H(bx) - F_{bg}(bx, t) \quad (4)$$

In both the plots shown in Figure 4, the black lines are the measured signal, $H(bx)$ and the black dots are the amplitudes from the multi-fit, A_i^{mf} . In the plot on the left the double dotted dashed lines show the nine OOT pulses, $F_i^{OOT}(A_i^{mf}, t)$, that are reconstructed from the amplitudes A_i^{mf} . In the plot on the right the dotted dashed

line is the background pulse shape, $F_{bg}(bx, t)$, as defined in Equation 3 and the dashed line is the recovered signal, $A^{sig}(bx, t)$ defined in Equation 4.

$$d(\delta) = (A \star T)(\delta) \equiv \frac{\sum A^{sig}(bx, t) S[bx, \delta]}{\sqrt{\sum (A^{sig}(bx, t)^2 \sum (S[bx, \delta])^2)}} \quad (5)$$

The time of the in-time pulse is determined by finding the time that best matches $A^{sig}(bx, t)$ to the pulse shape template $S[bx]$, where $S[bx]$ is a discrete function of the BX. This is done by constructing a distance, δ , between $A^{sig}(bx, t)$ and the pulse shape template, $S[bx]$, that has a minimum when there is a best overlap between the signal and the template. The time that maximizes the cross correlation function, $d(\delta)$ given in Equation 5, is then the reconstructed time for a rechit in a crystal. This second step is referred to as the cross correlation fit.

In the case where the reconstructed time is delayed more than ~ 1 ns, the OOT amplitude correction returns a biased result that tends to underestimate the reconstructed time by a factor of two. This effect is discussed in the next section, Section 2.2.1.

2.2.1 Delay Time Bias

The majority of signals searched for using ECAL time reconstruction involve “prompt” signals. Prompt signals are the result of a stable particle that interacts with the detectors that are produced with no significant displacement, or delay, from the time and location of the primary interaction (PV). There is a rich field of study involving searches for signals that are not prompt. These “delayed” signals often result when a semi-stable particle, often called a long lived particle (LLP), is produced and travels a significant distance from the PV, often at $\beta < 1$, before it decays resulting in the production of stable particles that are then seen in the detectors. The total time it takes for the LLP to decay plus the time for the resultant stable particles to travel to the detectors is typically larger than the time it would take for a stable particle to travel from the PV directly to the detectors. Such signals are said to have “large delay times”. The performance of the CC algorithm was studied with simulated samples where large delay times are expected due to the presence of LLPs.

In order to study the accuracy of the reconstructed time it is necessary to compare the reconstructed time to a “true”, or truth, time. A truth time for each reconstructed time is determined using the record of the amount and time of the energy deposits in each ECAL crystal present in the MC dataset, called pCaloHits. This truth time, or gen-time, for a rechit is calculated from the energy weighted average of the pCaloHit times used to simulate the pulse from which the rechit time is reconstructed.

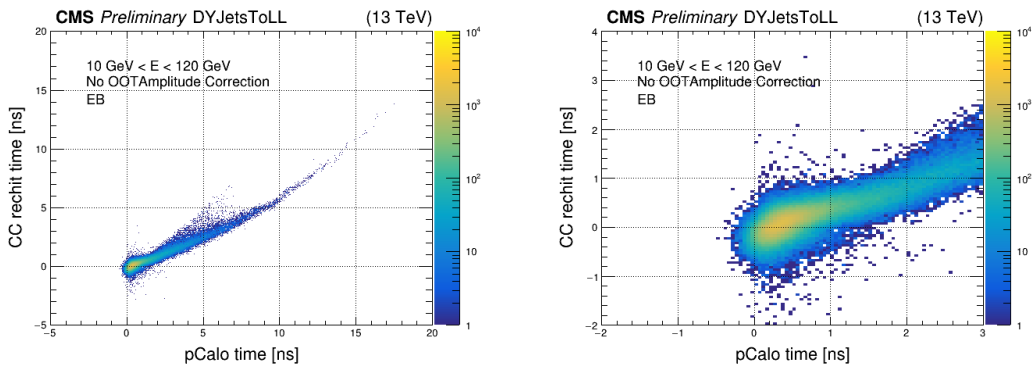


Figure 5: The gen-time for each rechit is plotted with respect to the rechit time produced by the CC algorithm in the HTTo2LongLivedTo4b Monte Carlo dataset. The gen-time is the energy weighted average of the gen-time associated with a particular rechit. The CC rechit time has no calibration applied. In the plot on the left a significant bias is shown in the reconstruction for times delayed more than ~ 1 ns. In the plot on the right, the scale is expanded to examine the performance of the KUCC algorithm for times less than ~ 1 ns, where it is seen that there is very good agreement with the gen-time.

In Figure 5 the gen-time for each rechit was determined and plotted against the rechit time produced by the CC algorithm. In the plot on the left of this figure a significant bias is shown in the reconstruction for times delayed

more than ~ 1 ns. In the right hand plot in this figure, the scale is expanded to examine the performance of the CC algorithm for times less than ~ 1 ns. At this scale it is seen that there is very good agreement between the CC reconstruction and the gen-times for prompt signals.

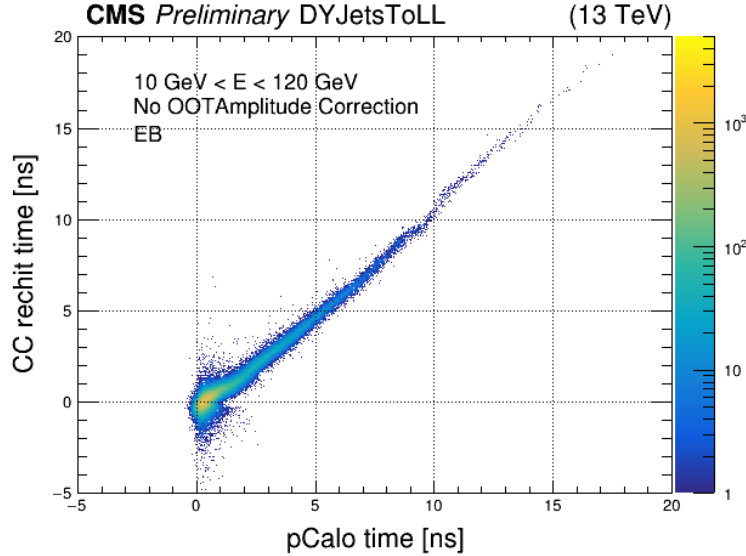


Figure 6: The gen-time for each rechit is plotted with respect to the rechit time produced by the CC algorithm, with the OOT amplitude corrections turned off in the time reconstruction. The CC rechit time has no calibration applied. Here we see that when the OOT amplitude corrections in the CC algorithm are disabled there is no bias at large delay times in the time reconstruction.

In Figure 6 the time reconstructed by a modified CC algorithm is plotted with respect to the gen-time. In this modified version of the CC algorithm, the OOT amplitude corrections to the pulse shape are not applied. In this plot no bias is observed at for large delay times, thereby demonstrating that the bias in the CC reconstruction is introduced by the OOT amplitude corrections. As discussed in Section 2.2, the OOT amplitude correction step utilizes the OOT amplitudes produced by the multifit algorithm to remove the contributions of OOT pulses in the measured pulse in order to recover the “in-time” pulse. When the OOT amplitudes are determined in the multifit algorithm, it is assumed that the OOT amplitudes occur at fixed times. Since OOT contributions to the pulse shape do not occur at fixed times, this introduces a bias in the OOT amplitudes.

To account for this bias the CC algorithm reconstructs the rechit time with and without the OOT amplitude correction. The time with the OOT amplitude correction is used for the time of the rechit. The difference between the time with and without the OOT amplitude correction is found and preserved. While the uncorrected CC times do not benefit from OOT-PU mitigation, the unbiased times are of interest for an analysis involving LLPs. The time reconstructed without the OOT amplitude correction can be by utilizing a dedicated function in the ECAL rechit collection of the CMS analysis framework (CMSSW).

2.2.2 Spike Rejection Parameters

Anomalous isolated signals with energies that can exceed 100 GeV have been observed in the ECAL. These deposits, called “spikes”, occur at a rate that is proportional to the collision rate. These “spikes” create issues for triggering at high luminosity and can cause large biases in the reconstructed energies and times of electrons, photons, jets, and in the computation of the missing transverse energy in an event. Spikes result from particles striking the APDs in the ECAL barrel directly and occasionally cause large anomalous signals through direct ionization of the silicon.[12] Spikes predominately deposit energy in a single crystal and are characterized by a “spike” like pulse shape with a fast rise-time and no scintillation component. Typically there is little or no activity in the crystals surrounding the crystal in which the spike occurs, though occasionally a “double” spike does occur where spike-like signals are seen in two adjacent crystals. A rechit that is produced by a spike in a crystal will be referred to as a “spike rechit”. The performance of traditional flags to reject spike rechits were studied when used

with the CC algorithm. The rechit collections for both the ratio and the CC algorithms were produced and stored without any downstream cleaning, filtering, or calibration for this study. The CC reconstruction was performed without OOT amplitude corrections to minimize the impact of the energy dependence on the time reconstruction. Such collections will be referred to as “raw” rechit collections.

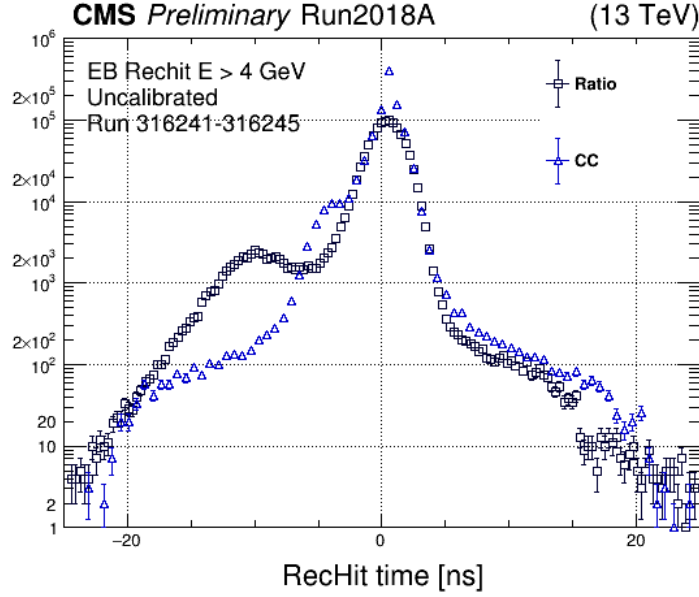


Figure 7: The distribution of the rechit times reconstructed using the ratio algorithm and the CC algorithm are plotted. The distributions were plotted for a rechits with energy > 4 GeV. Full rechit collections are produced for both the ratio and CC reconstruction. These collections were saved as output by the producers without downstream cleaning, filtering, or calibrations. The bumps in the distributions located at roughly -10 ns in the distribution for the ratio method and -5 ns for the CC method correspond to spike rechits.

In Figure 7 the distribution of the rechit times in the raw collections for both the CC and ratio method is shown for rechits in which the rechit energy is > 4 GeV. The bumps located at roughly -10 ns in the distribution for the ratio method and -5 ns for the CC method correspond to rechits resulting from spikes. In Figure 8 the raw rechit times produced with the CC algorithm are plotted with respect to the raw rechit times produced with the ratio algorithm for rechits with energies between 10 GeV and 120 GeV. In this plot we can see that there is a strong correlation between the reconstructed times for the two methods. It can also be seen that the CC method produces a much narrower spread for prompt times than the ratio method. The “spike” rechit population is located between -6 ns and -15 ns on the ratio time axis and between -2 ns and -10 ns on the CC time axis.

As was seen in Figure 7 and Figure 8, it is characteristic of a spike rechit that the spike-like pulse shape results in a reconstruction with a time that is significantly earlier than for a non-spike rechit. The fast rise-time of the uncharacteristic shape of a spike signal causes the ratio algorithm to reconstruct a significantly earlier time for spike rechits. The CC algorithm is less sensitive to the bias introduced by the fast rise time and reconstructs a time that is less significantly early but still earlier than non-spike rechits. It is also characteristic of a spike rechit that they are isolated from other rechits. Since a crystal with a large energy deposition that is isolated from other crystals with activity is inconsistent with an electromagnetic shower, an isolated rechit would not be expected for non-spike rechits. A spike rechit can then be identified by looking at both the isolation of the rechit and the reconstructed time of the rechit.

The swiss-cross variable characterizes the energy sharing between a crystal with a high energy signal and its neighboring crystals. The swiss-cross can be thought of as a ratio where the more isolated a rechit is the closer the swiss-cross value is to one. The swiss-cross is used to define two spike rechit rejection flags. The kWeird flag

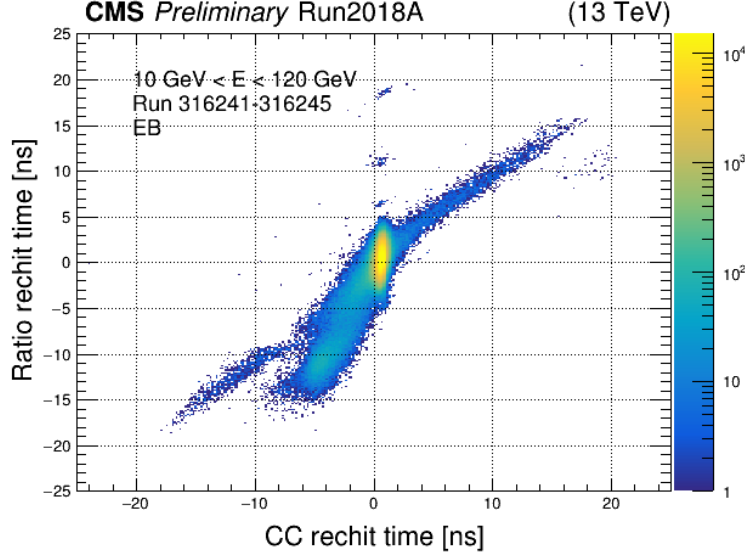


Figure 8: The times for raw rechit collections for the CC algorithm and the ratio algorithm is plot with respect to each other for rechit energies between 10 GeV and 120 GeV. It can be seen that there is a strong correlation between the reconstructed times for the two methods and that the CC method produces a much narrower spread for prompt times than the ratio method. The “spike” rechits found in the population located between -6 ns and -15 ns for ratio algorithm and between -2 ns and -10 ns for KUCC algorithm.

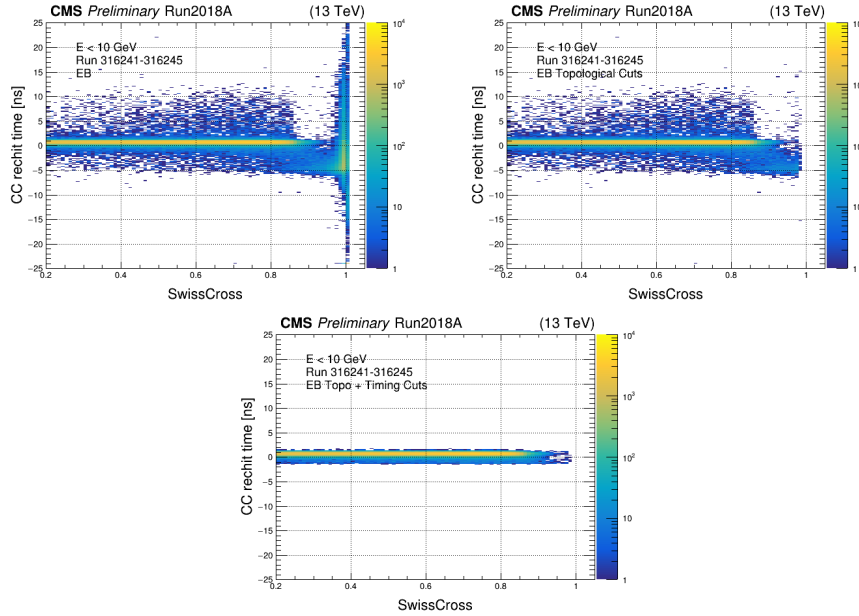


Figure 9: Swiss-cross is plotted with respect to the CC reconstructed time for rechits with energy > 10 GeV. In the plot on the top left the full distribution is shown with no spike flag cuts applied. As can be seen, this plot is composed of two populations, “non-spike” rechits spread around a time of 0 ns with a swiss-cross > 0.95 and “spike” rechits spread around a swiss-cross of 1.0 with a “bump” located roughly at a time of -5.0 ns. In the plot on the right the topological flags are applied as cuts to reject spike rechits. With these cuts most of the “spike” rechits are rejected with the exception of “bump” located roughly at -5.0 ns. In the bottom plot the kOOT flag cut is also applied in addition to the topological cuts. For the kOOT flag a value of 3 ns is used for all threshold parameters. Here it is seen that all “spike” rechits are rejected.

that rejects rechits with isolated energy depositions, and the kDiWeird flag that is designed to reject double spikes.[12] The use of the combination of kWierd or kDiWierd is referred to as topological (topo) cuts. In the time reconstruction algorithm a spike rechit rejection flag called kOutOfTime (kOOT) is defined based on energy dependent time thresholds that can be set independently for the barrel, the endcaps, and both the high and low energy reconstruction regions. This threshold defines a “window” around the “prompt” times and flags any times that fall outside this window in both the early and delayed sides of the time distribution.

In Figure 9 the swiss-cross is plotted with respect to the CC reconstructed time for rechits with energy > 10 GeV. In the plot on the top left the full distribution with no spike rechit cuts applied. As can be seen, this plot is composed of two populations, “non-spike” rechits spread around a time of 0 ns with a swiss-cross > 0.95 and “spike” rechits spread around a swiss-cross of 1.0 with a “bump” located roughly at a time of -5.0 ns. In the plot on the right, the topological cuts are applied. With these cuts most of the “spike” rechits are rejected with the exception of “bump” located roughly at -5.0 ns. In the bottom plot the kOOT flag is applied in addition to the topological cuts. In this plot a threshold value of 3 ns is used for all kOOT threshold parameters as is the default in the ratio algorithm. In this plot it can be seen that the majority of “spike like” rechits are rejected.

If a rechit is flagged as kOOT, that rechit is much less likely to be utilized in down stream particle reconstruction and thereby impacting the energy of these reconstructed particles. In order to avoid adverse impact on the energies of the reconstructed particles it was necessary to carefully tune the thresholds used to set the kOOT flag with the CC time reconstruction. Since times reconstructed with the CC algorithm depends on the amplitudes it a significantly different manner then the ratio reconstruction, the thresholds for low and high energies tuned independently. Appropriate thresholds were determined in order to maximize spike rechit rejection, as seen in Figure 9 , without negatively impacting the down stream particle reconstruction, across all reconstructed rechit energies.

3 ECAL Timing Performance

3.1 Resolution

Resolution studies are used to characterize the precision of a timing reconstruction method. The time resolution is determined by the difference in the times of two recorded hits (rechit) in a crystal from a pair of crystals in which the rechits should have been produced simultaneously. As shown in Figure 10, two methods of selecting this pair of crystals are used. The first method selects for a pair of adjacent crystals that have similar energies to measure a “local resolution”. The “local resolution” is used to characterize the single crystal time resolution with, or without, the effects of delays due to the read out unit electronics. The second method selects for a pair of rechits that correspond to hits generated by electrons from a $Z \rightarrow e^+e^-$ process. This second method selects for a “global resolution” which is used to characterize time resolutions typically obtained in physics analyses.

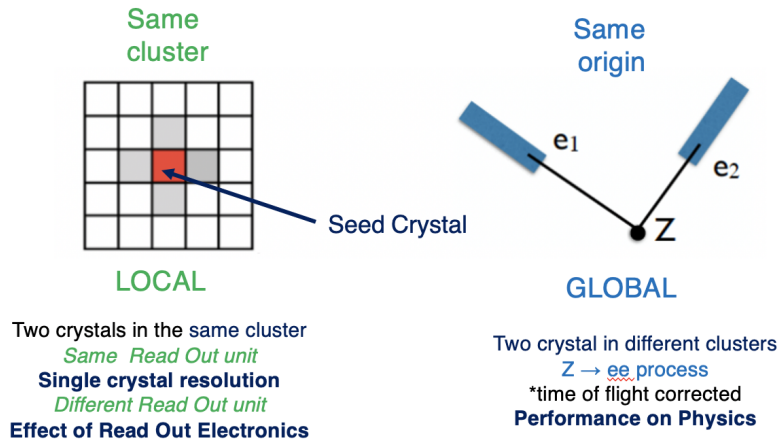


Figure 10: The two methods of selecting pairs of crystals to use in determining the time resolution. In a local resolution, used to characterize the single crystal time resolution and the effects of delays due to the read out unit electronics, two adjacent crystals in the same cluster are selected. For a global resolution, used to characterize time resolutions typically obtained in physics analyses, two crystals in different clusters that corresponded to the electrons from a $Z \rightarrow e^+e^-$ process are selected.

For a local resolution, shown on the right in Figure 10, rechits for a pair of crystals from the cluster of crystal that form an EG candidate are selected. The pair of crystal are required to be the seed crystal of the EG candidate and an adjacent crystal in either the η or ϕ directions for which the rechit energy of the second crystal is required to be within 20% of the energy of the rechit in the seed crystal. The selected EG candidates from which the crystal pair is taken is required to have the semi-minor axis of the second moment of the crystal cluster geometrical distribution (smin) to be < 0.3 , and the semi-minor axis (smaj) to be < 0.5 . When more than one EG candidate has a qualifying crystal pair in a collision event, the EG candidate with the largest momentum in the event is selected. Local resolutions are further divided into a case where both crystals share the same readout electronics, referred to as a 'Same RO Unit' resolution, and a case where the crystals in the pair have a different readout unit, referred to as 'Different RO Unit' resolution, depending on whether the two crystals are located in the same trigger tower or if they are in adjacent trigger towers.

In a global resolution, shown on the left in Figure 10, the seed crystals from two EG candidates that match electrons from $Z \rightarrow e^+e^-$ decays are selected. The EG candidates are required to be matched to electron candidates with $dR < 0.1$ where the electron candidates track z is within 0.005 cm of the primary vertex z for the event. The pair of EG candidates that have an invariant final state mass between 60 and 150 GeV are selected. If more than one pair is found in an event, the pair with final state mass nearest to the Z mass is selected. For each of the two selected EG candidates, the time-of-flight (TOF) from the primary vertex (PV) of the event, where the Z decay is assumed to have occurred, to the physical location seed crystal of the EG candidate is typically not the same. To account for this a correction for the TOF of two EG candidates is applied to the rechit times. A global resolution is referred to as " $Z \rightarrow e^+e^-$ " resolution. The selection requirements for local and global resolutions are summarized in Table 1.

As shown in Figure 11, the difference in the rechit times for a pair of selected crystals, $\Delta(SeedTime)$, is plotted as a function of the "effective amplitude" of the rechit amplitudes. The "effective amplitude" of the rechit amplitudes is proportional to the energy of the corresponding EG candidates.

$$A_{eff}/\sigma_N = \frac{(A_1/\sigma_{N_1})(A_2/\sigma_{N_2})}{\sqrt{(A_1/\sigma_{N_1})^2 + (A_2/\sigma_{N_2})^2}} \quad (6)$$

The effective amplitude, A_{eff}/σ_N , is defined in Equation 6 where A_i is the amplitude of the signal detected by crystal i and σ_{N_i} is the pedestal noise for that crystal. The distribution of $\Delta(SeedTime)$ in each bin of A_{eff}/σ_N is fitted to a Gaussian function. The standard deviation, $\sigma(t_1 - t_2)$, of each of $\Delta(SeedTime)$ distributions in each bin A_{eff}/σ_N is plotted as a function of the central value of the A_{eff}/σ_N bin.

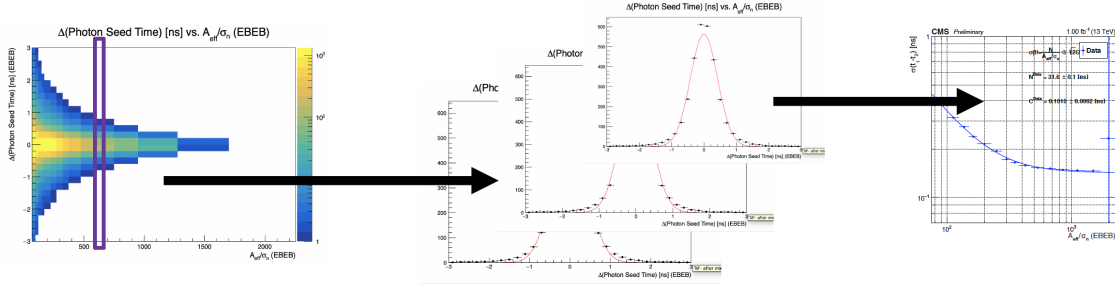


Figure 11: To measure the time resolution the $\Delta(SeedTime)$ is plotted in bins of A_{eff}/σ_N . The distribution of $\Delta(SeedTime)$ in each bin is plotted and fit to a Gaussian to extract the standard deviation, $\sigma(t_1 - t_2)$, of the distribution. The $\sigma(t_1 - t_2)$ for the distribution in each bin is plotted in bins that are the same as the bin from which it was derived. The resulting plot is fitted as a function of A_{eff}/σ_N as shown in Equation 5.

$$\sigma_i^2 = \left(\frac{N}{A_{eff}/\sigma_N}\right)^2 + 2C^2 \quad (7)$$

The dependence of $\sigma(t_1 - t_2)$ on A_{eff}/σ_N characterizes the time resolution. This dependence is traditionally fitted with the resolution function in Equation 7 for $A_{eff}/\sigma_N > 75$. In this equation the noise term (N) is the coefficient for the uncertainty due to noise and the constant term (C) has several contributions including the variability of the point of shower initiation within the crystal, systematic effects such as those due to small differences in pulse

shapes in different crystals, differences in processing delays due to different readout electronics, and pileup effects. The constant term is the resolution of the time reconstruction for the particular case (same read-out unit, different read-out unit, $Z \rightarrow e^+e^-$) considered.

$$\sigma_i^2 = \left(\frac{N}{A_{eff}/\sigma_N} \right)^2 + \frac{S^2}{A_{eff}/\sigma_N} + 2C^2 \quad (8)$$

As shown in Equation 8 the parameterization for the resolution can be extended with the inclusion of a stochastic term (S). While a stochastic term is not traditionally used, the inclusion of this term improved the fits when the limit A_{eff}/σ_N is lowered to 10 to better characterize the shape of the low energy dependance. This approach is taken for the resolutions of the time reconstruction using the CC method.

Resolution Selection Cuts
Local Resolution
Photon smin > 0.3 Photon smaj > 0.5 Photon seed crystal has an adjacent crystal where : seed crystal energy < 120% adjacent crystal energy Qualifying seed crystal with largest energy
Global Resolution
Photon has PixSeed Photon gedID < 3 Photon matches to an electron with dR < 0.1 Matching electron has trackZ within 1.0 cm of primary vertex Z At least 2 qualifying photons Pair of photons with combined mass < 150 GeV & > 60 GeV Qualifying pair of photons with combined mass closets to Z mass

Table 1: Selection cuts for local and global resolutions.

3.2 Calibration

In order to compare various timing algorithms it is necessary to calibrate the rechit times produced by the different timing algorithms in the same manner. Since, during development in Run 2, it is not possible to run the default timing calibration present in CMSSW on the CC algorithm, an external calibration based on the average rechit time by channel was implemented. Rechits selected for the channel by channel average rechit time are required to have a minimum energy to reduce the effects of noise. As shown in Figure 12, several different values for this energy cut were studied with the default CMSSW time reconstruction method in selected runs from the 2018D era. It was found that a minimum energy cut above 5 GeV produced a negligible improvement in the resolution when the corresponding calibration is applied to the rechit times and a minimum energy threshold of 5 GeV are used for the external calibration. This external calibration is always used for the CC method prior to the implementation of a CC based calibration in CMSSW during Run 3. An external calibration based on times reconstructed with the ratio method is also used as an additional external calibration to the ratio method times in all cases where the two timing methods are compared and an external calibration was required for the CC method.

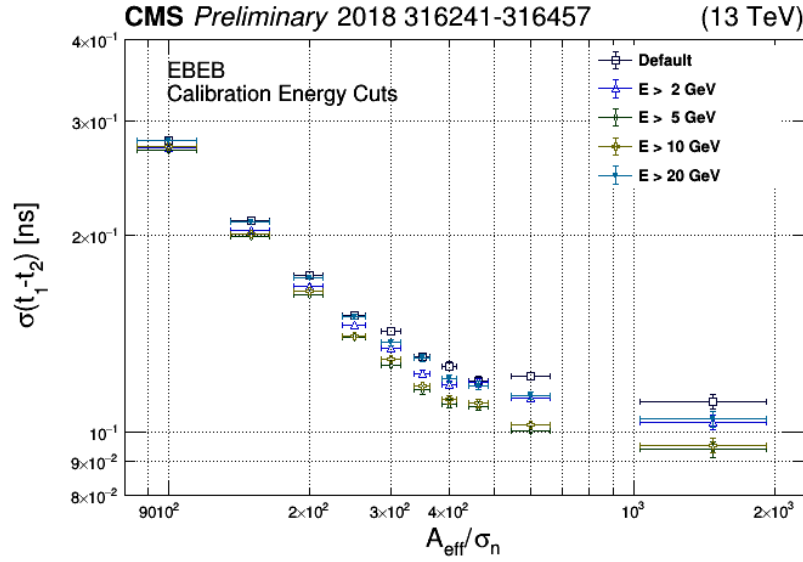


Figure 12: Same RO resolution with external calibration applied to default CMSSW timing reconstruction. The external calibration is calculated by taking the average rechit time by channel for rechits that have a minimum energy of 5 GeV over all events in the runs for which the calibration is used.

3.3 Performance of Cross Correlation Algorithm in Data

OOT-PU occurs when interactions from particles originating in events from BXs before the current BX are seen in events recorded during the current BX. In particular, signals produced by scintillating light pulses in ECAL crystals are longer than 25 ns, the nominal time between BXs. This can easily result in signals produced by particles originating in different BXs overlapping and impacting the time reconstruction. The effect of OOT-PU on the time reconstruction can be seen when the mean rechit time by BX position in the “train” of filled BXs is studied.

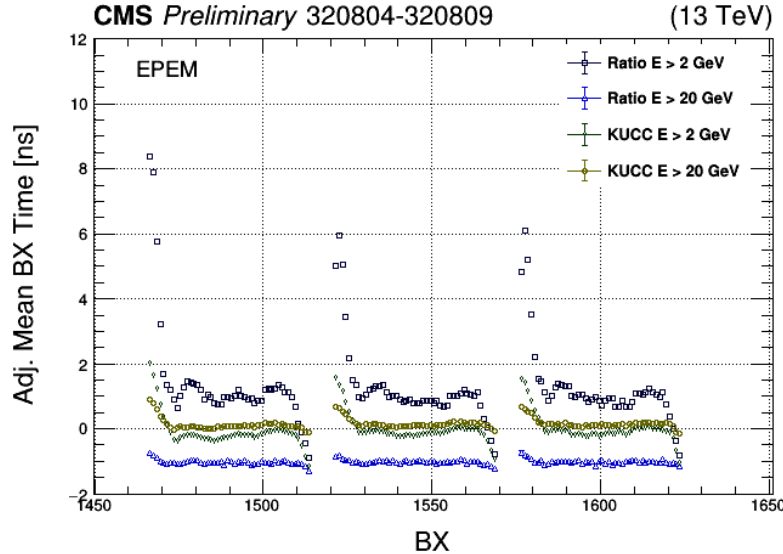


Figure 13: Mean rechit time by BX, for rechits with a minimum of 2 GeV (20 GeV) with both the ratio and CC reconstructions, for a typical train of filled buckets is plotted. The plotted values are adjusted so that the mean rechit time by BX when averaged over all BXs and energies is zero. The CC algorithm is shown to mitigate the impact of OOT-PU through the reduced dependence of the time on the BX position and to achieve a greatly reduced energy dependence.

In Figure 13 the mean rechit time by BX for rechits with a minimum of 2 GeV, and for rechits with a minimum of 20 GeV, for rechit collections produced with the default ratio algorithm and the CC algorithm are plotted. For this plot the rechit times are calibrated by subtracting the mean time across all BXs for rechits with a minimum of 5 GeV in the corresponding rechit collection. It can be seen in the leading and trailing tails of the trains that the CC algorithm significantly reduces dependence of the rechit time on the BX position and therefore mitigates the impact of OOT-PU on the reconstruction. The central regions of the sub-trains exhibit a markedly reduce spread between the 2 GeV and 20 GeV data points for the CC algorithm compared to the ratio algorithm indicating that the CC algorithm produces a greatly reduce energy dependence in the time reconstruction.

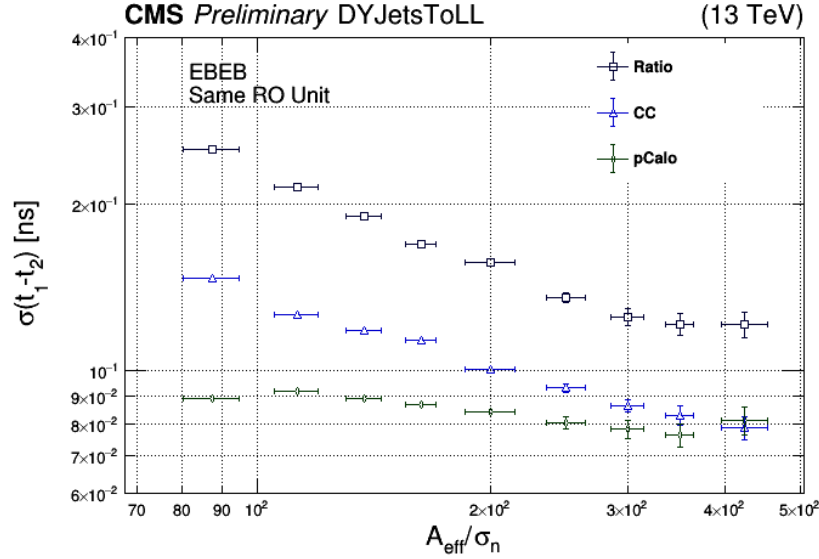


Figure 14: Same RO unit, single crystal resolutions for the ratio algorithm, the CC algorithm, and pCaloHit gen time are compared in simulation. The CC algorithm shows a much improved resolution compared to the ratio algorithm and approaches the pCaloHit gen time resolution at high energies.

During development, time using the CC algorithm for the time reconstruction were produced using a two tier processing approach. This allows for both the time produced with the CC time reconstruction and the already existing time from the ratio time reconstruction to be recorded for each rechit. With this information the performance of the two algorithms can be directly compared. Studies comparing the performance of the CC and ratio time reconstruction were done in both simulation (MC) and recorded data.

The resolution obtained by the CC algorithm in MC were studied in comparison to both the ratio algorithm and to pCalo gen times. these studies were performed with Run 2 MC using the two tier data processing approach and an external calibration. Since, as noted in section ??, for MC the data used as input for the timing reconstruction methods is derived from the pCaloHit collection. Therefore, the resolution obtained using the pCaloHit gen time reference is a reasonable estimate of the best resolution achievable by a timing reconstruction method. In Figure 14 the single crystal resolutions for the ratio algorithm, the CC algorithm, and pCalo gen time are plotted. It is seen here that the resolution of the CC algorithm is significantly narrower than the resolution of the ratio algorithm across all effective amplitudes and at larger effective amplitudes the resolution of the CC algorithm approaches the pCalo gen time resolution.

Figure 15 shows the same readout unit and $Z \rightarrow e^+e^-$ resolution in Run 2 using the two tier data processing approach and external calibrations for recorded data. The CC algorithm produces an improvement ~ 30 ps for the same RO unit resolution and an improvement of ~ 35 ps for the $Z \rightarrow e^+e^-$ resolution compared to the ratio algorithm reconstruction. In Figure 16 resolutions obtained in Run 3 are studied. During Run 3 the CC algorithm was implemented in CMSSW and a dedicated online calibration was implemented. This plot shows the comparative resolutions obtained from two separate full prompt reconstructions of the same data where the ratio

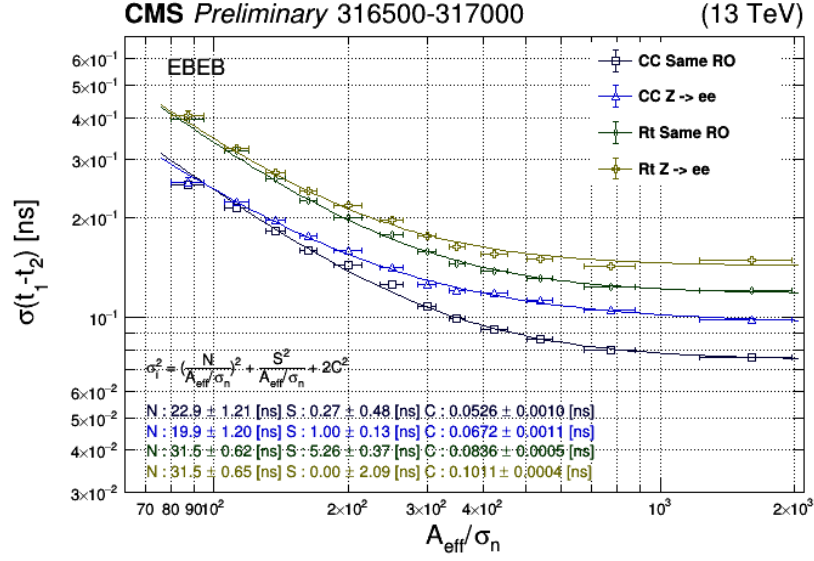


Figure 15: Comparison of time Same RO and $Z \rightarrow e^+e^-$ resolution for CC and the ratio algorithms in Run 2 using the two tier data processing approach and external calibrations. This plot shows that the CC algorithms achieves an improvement in the Same RO unit resolution of ~ 30 ps and an $Z \rightarrow e^+e^-$ resolution improvement of ~ 35 ps.

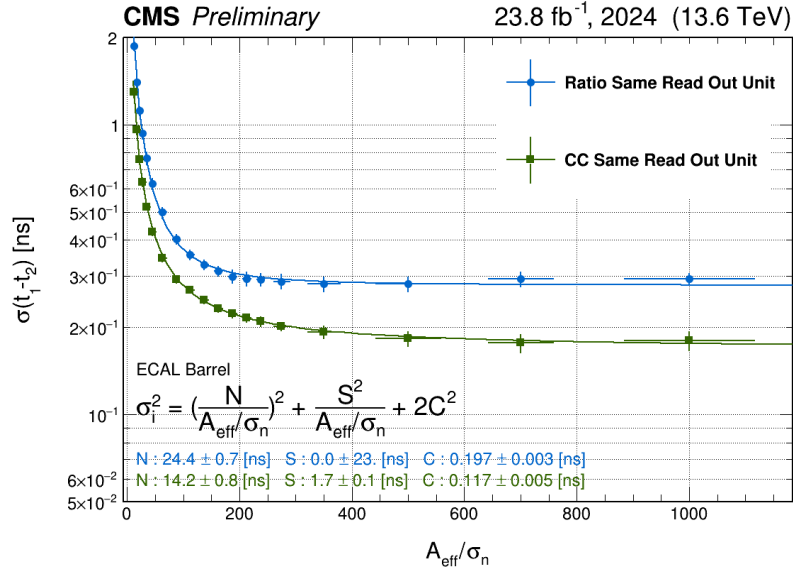


Figure 16: Comparison of time Same RO and $Z \rightarrow e^+e^-$ resolution for CC and the ratio algorithms in Run 3. In Run 3 the CC algorithm had been implemented in CMSSW and a dedicated online calibration was implemented. This plot shows the comparative resolutions obtained from two separate full prompt reconstructions of the same data where the ratio was default for one of the reconstructions and the CC method was the default for the other reconstruction. This plot shows that the CC algorithms achieves an improvement in the Same RO unit resolution of ~ 80 ps.

was default for one of the reconstructions and the CC method was the default for the other reconstruction. Here we see that the CC algorithms achieves an improvement in the Same RO unit resolution of ~ 80 ps. The CC time reconstruction was the default time reconstruction used for data taking in 2025 of Run 3. In Figure 17 we see the same readout unit and $Z \rightarrow e^+e^-$ resolutions obtained by the CC reconstruction during the first data taking period with sufficient statistics for a resolution study in 2025. The CC algorithms achieves a Same RO unit resolution of ~ 135 ps and a $Z \rightarrow e^+e^-$ resolution of ~ 180 ps. Comparing the $Z \rightarrow e^+e^-$ resolution of ~ 180 ps achieved by the CC time reconstruction to the Same RO unit resolution of ~ 200 ps achieved by the ratio time reconstruction, as seen in Figure 16, we can deduce that the CC time reconstruction significantly mitigates the impact of OOT-PU on the time reconstruction.

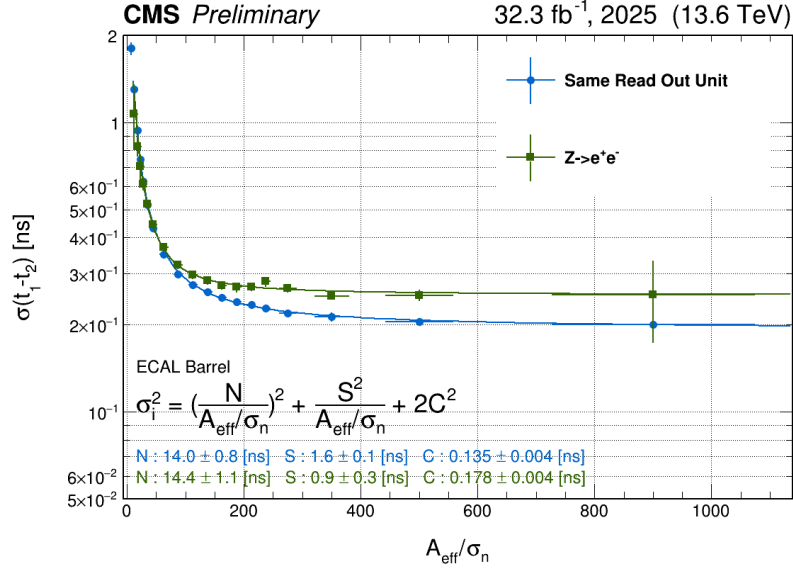


Figure 17: Same RO and $Z \rightarrow e^+e^-$ resolution for CC in Run 3. The CC algorithm, along with a dedicated online calibration, was implemented in CMSSW in 2024 of Run 3 and was the default ECAL time reconstruction in 2025. This plot shows the Same RO and $Z \rightarrow e^+e^-$ resolutions for obtained during a prompt reconstruction with the CC method. Here we see that the CC algorithms achieves a Same RO unit resolution of ~ 135 ps and an $Z \rightarrow e^+e^-$ resolution of ~ 180 ps in prompt reconstruction.

4 Summary

In the HL-LHC era the expected instantaneous luminosities will be a factor of five larger than the current LHC nominal conditions. While these more intense conditions will enable the LHC experiments to accumulate datasets more than an order of magnitude larger than all of the preceding LHC runs, it will also exacerbate the experimental challenge of the multiple pile-up interactions appearing in each event. To mitigate degradation of LHC event reconstruction performance resulting from potentially hundreds of pile-up (PU) interactions expected in an HL-LHC event, both CMS and ATLAS will incorporate precision timing measurements into their detectors, significantly expanding the capabilities of the experiments through a combination of new sub-detectors and dedicated upgrades to existing detectors, such as the ECAL. This will facilitate the development of 4D reconstruction methods that will provide new tools for both pile-up suppression and a range of new experimental techniques to probe physics signatures and long-lived particles.

The ECAL, through a combination of the scintillation timescale of PbWO_4 crystals, the electronic pulse shaping, and the sampling rate, allows for excellent time resolution to be obtained. The time reconstruction in ECAL is sensitive to the presence of pulses arising from particles produced in PU collisions. As the LHC moves to higher luminosity in the HL-LHC era, the effects of OOT-PU in the events become a significant issue in many areas. The ECAL performance group has set a goal to achieve a ~ 30 ps time resolution for HL-LHC data taking. As a significant step to prepare for operations in high OOT-PU environments, and to meet this goal, studies of methods

to mitigate the effects of OOT-PU on the timing reconstruction have been performed. Development of a ECAL timing reconstruction method that is OOT-PU aware will result in an improved time resolution that will not only improve the precision of many current physics measurements, it will also provide a foundation to meet the goal of the ECAL to achieve a ~ 30 ps time resolution. The Cross Correlation time reconstruction (CC) was developed to meet these challenges. The CC time reconstruction will not only support the incorporation of precision timing in the HL-LHC era, but will also allow us to begin to explore some of the possibilities of 4D reconstruction methods, that will be facilitated by precision timing, in Run III.

The CC time reconstruction leverages the existing OOT pulse amplitudes determined by the multi-fit time reconstruction for the timing reconstruction. Given these OOT amplitudes and the known pulse shape of an isolated signal in a crystal, the nominal, in-time, pulse contribution in the measured amplitudes for a rechit is reconstructed. By extracting the nominal pulse to use in reconstruction of the rechit time and finding the time that, via a cross correlation function, best matches the extracted signal amplitudes to the pulse shape template, the effects of OOT-PU are mitigated, significantly improving the ECAL time resolution.

In studies of average rechit time by BX, the CC time reconstruction significantly reduces dependence of the rechit time on the BX position, mitigating the impact of OOT-PU on the reconstruction, and greatly reduces the energy dependence in the time reconstruction. In direct comparison, the resolution of the times reconstructed by the CC time reconstruction in data show that the CC time reconstruction produces an improvement ~ 30 ps for a same RO unit resolution and an improvement of ~ 35 ps for the $Z \rightarrow e^+e^-$ resolution compared to the default ratio time reconstruction in Run 2. In Run 3 the CC time reconstruction was implemented in CMSSW and an online time calibration for CC was deployed. A devoted full data reconstruction where CC was used as the default ECAL time reconstruction was produced for a selected data taking period and time resolution achieved was compared to the nominal prompt reconstruction for the data period. In these datasets it was seen that CC achieved an improvement of ~ 80 ps for the same RO unit resolution over the ratio time reconstruction. In year 2025 of Run 3 the CC time reconstruction was deployed as the default time reconstruction for data taking. The CC time reconstruction has achieved SRU resolution of ~ 135 ps and $Z \rightarrow e^+e^-$ resolution of ~ 180 ps in prompt reconstruction in early 2025 data. With the utilization of the CC time reconstruction we expect comparable, if not improved, resolutions for all of Run III.

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